FML · Unit 9 · Supplement

Choosing ε and Perplexity

A short beginner's guide to the two knobs you must set yourself

k-distance plot → Hand example → Why the bend → Better rules → Perplexity

DBSCAN and t-SNE each leave one knob for you to set

Everything else is computed from the data

DBSCAN · ε

How far to look

the circle drawn around each point

DBSCAN · minPts

How many make a crowd

we use 4, counting the point itself

t-SNE · PERPLEXITY

How many friends

each point keeps about this many

This supplement shows how to pick ε first, then perplexity.

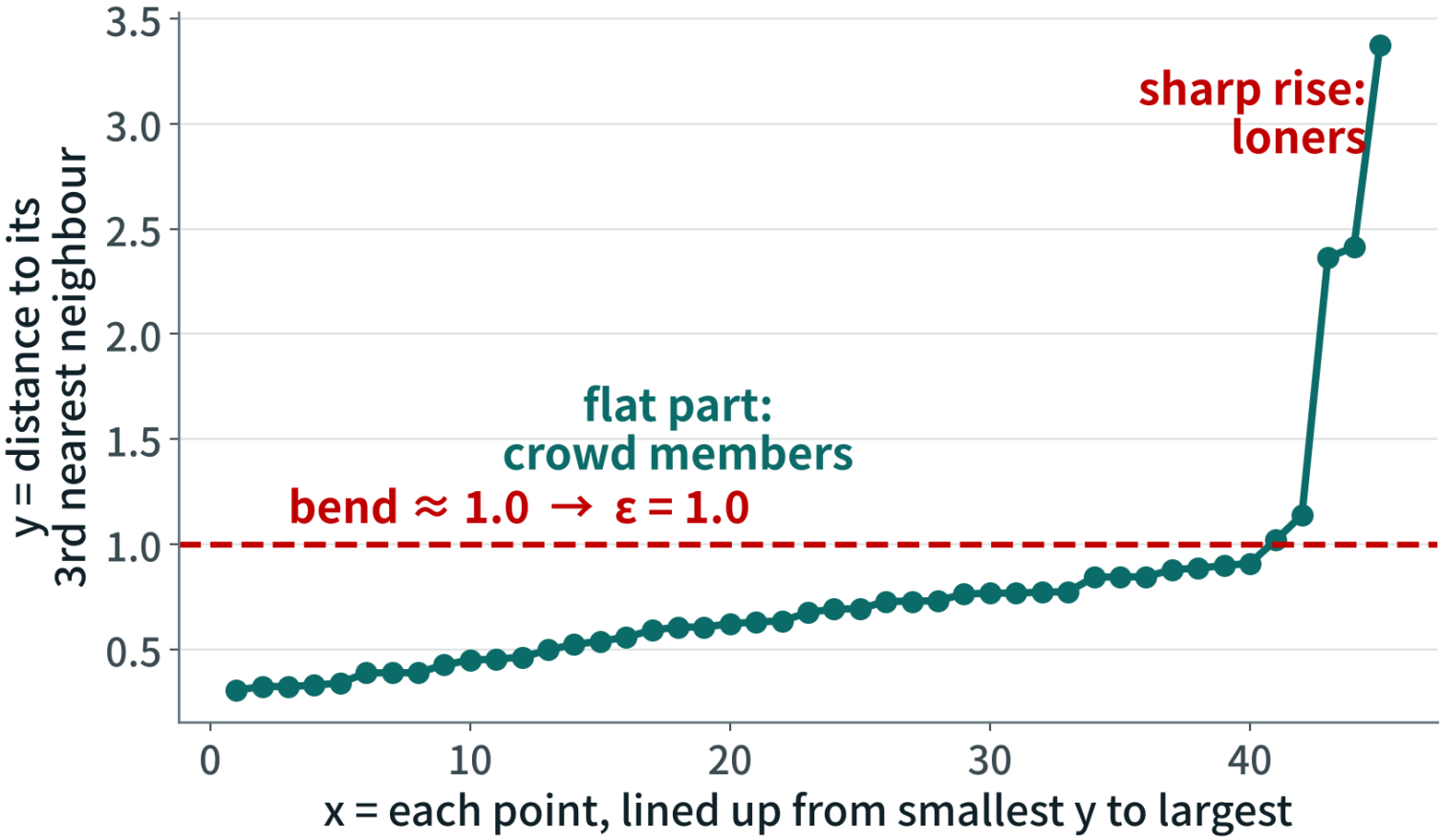
The k-distance plot shows how far each point must reach

minPts = 4 counts the point itself, so we look at the 3rd nearest neighbour

Read the axes

- x = a rank only
- y = one point's value
- Not an average
- Not a running total

Pick ε at the bend



Six points on a line make a hand example

A to E sit close together; F sits far away. minPts = 4, so each point needs 3 neighbours



Step 1: list each point's distances and take the 3rd one

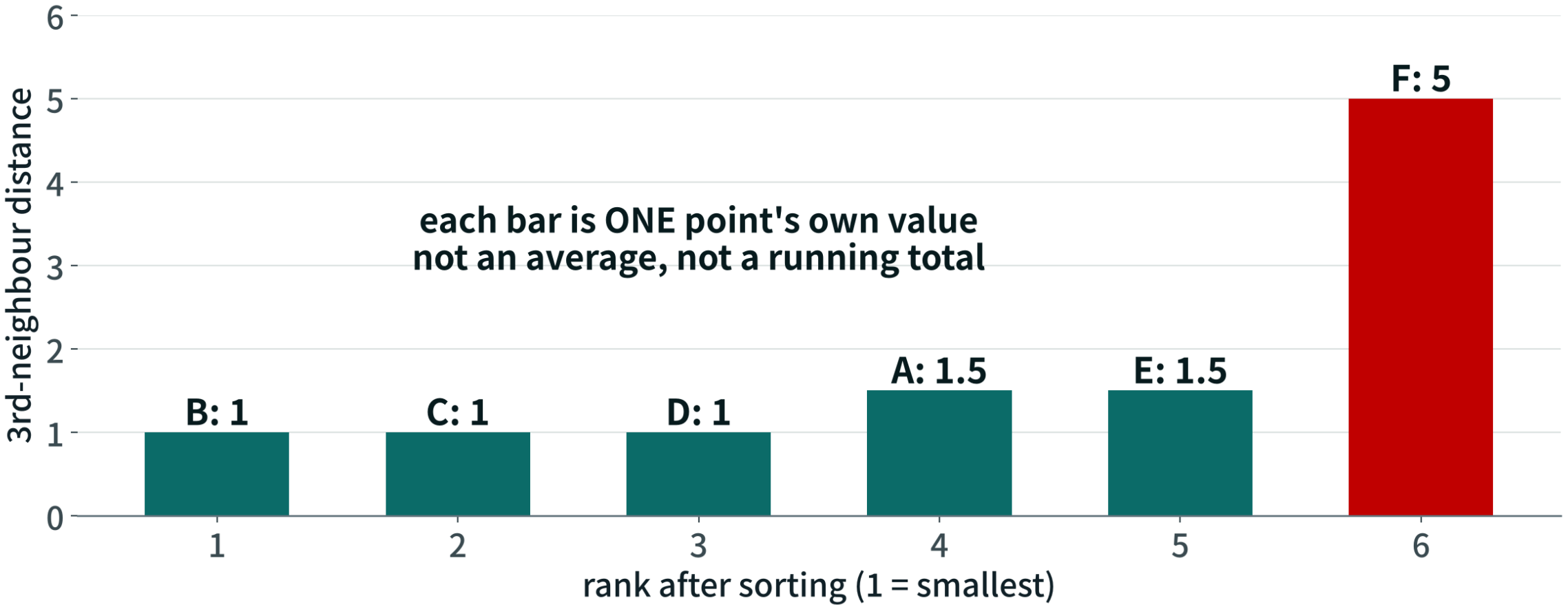
Distances to the other five points, nearest first

Point	Distances to the others (near → far)	3rd
A (0)	0.5, 1.0, 1.5, 2.0, 6.0	1.5
B (0.5)	0.5, 0.5, 1.0, 1.5, 5.5	1.0
C (1.0)	0.5, 0.5, 1.0, 1.0, 5.0	1.0
D (1.5)	0.5, 0.5, 1.0, 1.5, 4.5	1.0
E (2.0)	0.5, 1.0, 1.5, 2.0, 4.0	1.5
F (6.0)	4.0, 4.5, 5.0, 5.5, 6.0	5.0

Example: B has A and C at 0.5 and D at 1.0, so its 3rd distance is 1.0.

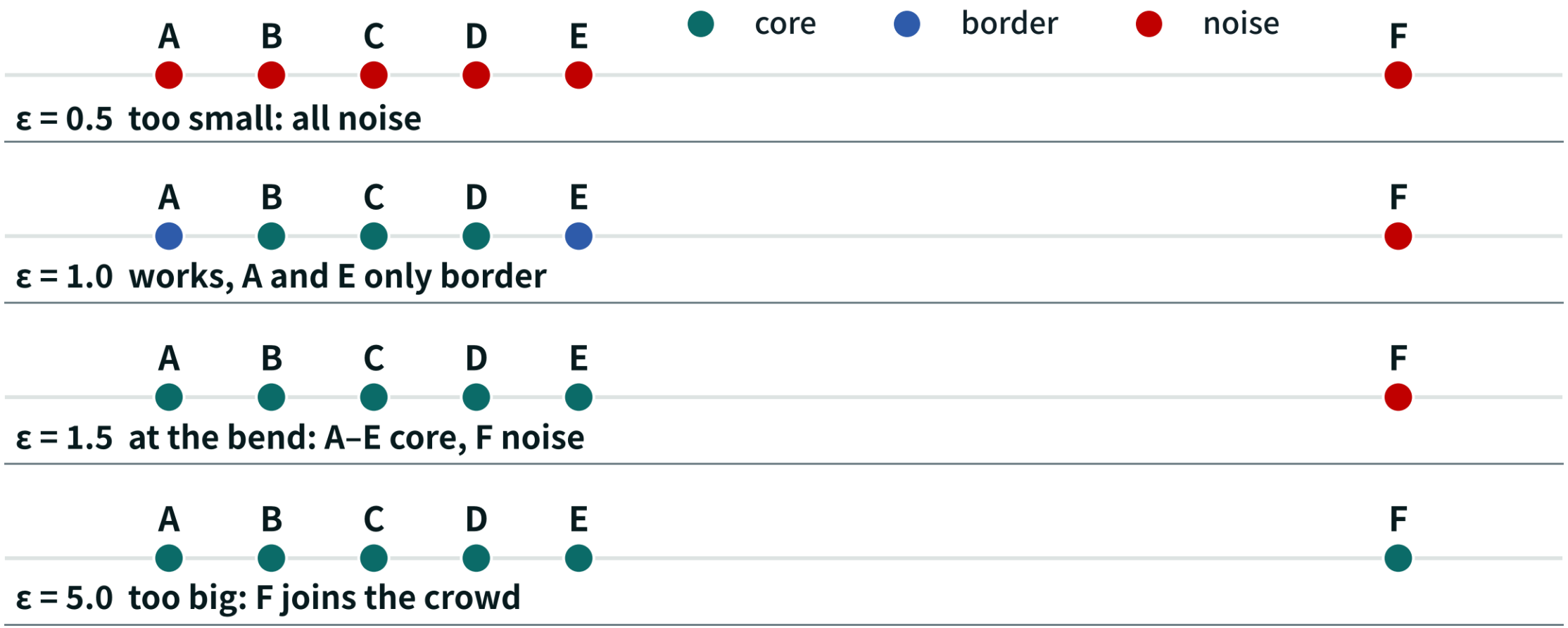
Step 2: sorting the six values makes the jump at F visible

This bar chart is the k-distance plot for six points



The bend is the smallest ε that keeps the crowd and drops the loner

The same six points labelled by DBSCAN at four values of ε



Below the bend crowds break up; above it loners join in

Three outcomes from the hand example

TOO SMALL · ε = 0.5

Crowd breaks

nobody gets 3 neighbours, all noise

AT THE BEND · ε = 1.5

Just right

A–E core, F noise

TOO BIG · ε = 5.0

Loner joins

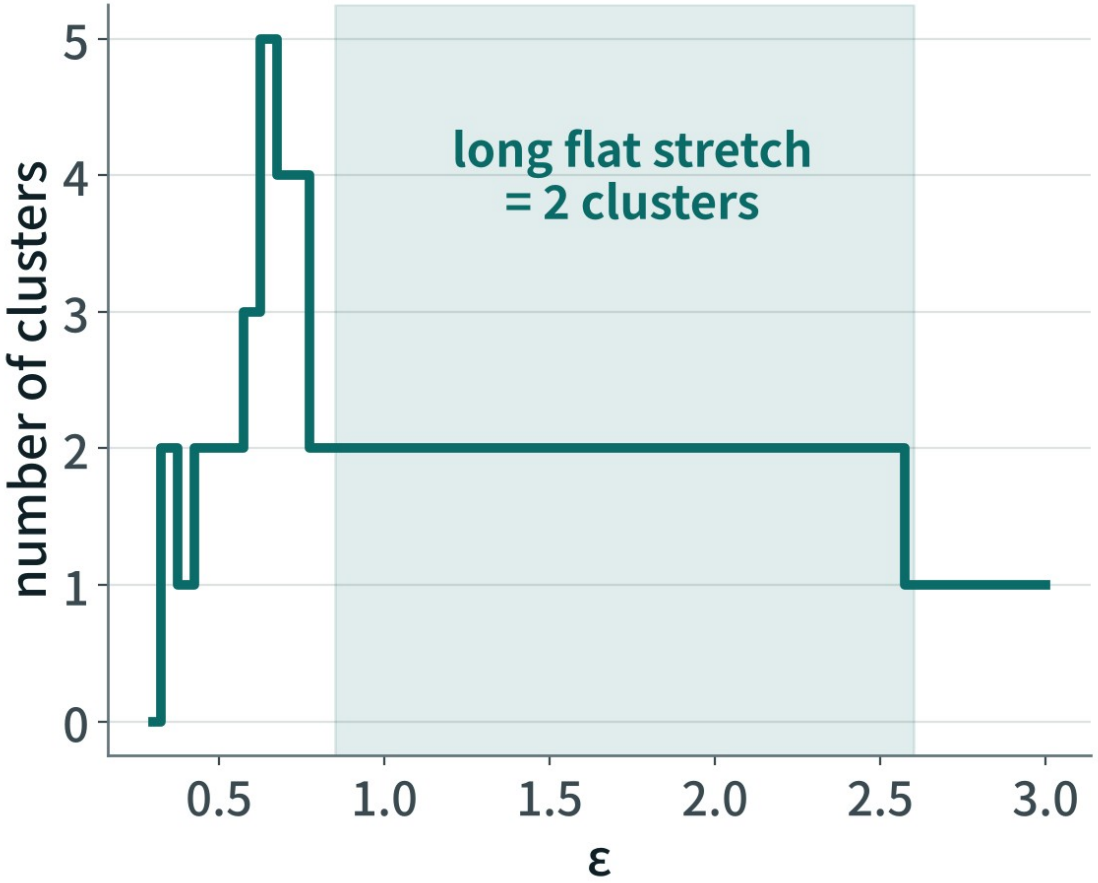
F gets 3 neighbours too

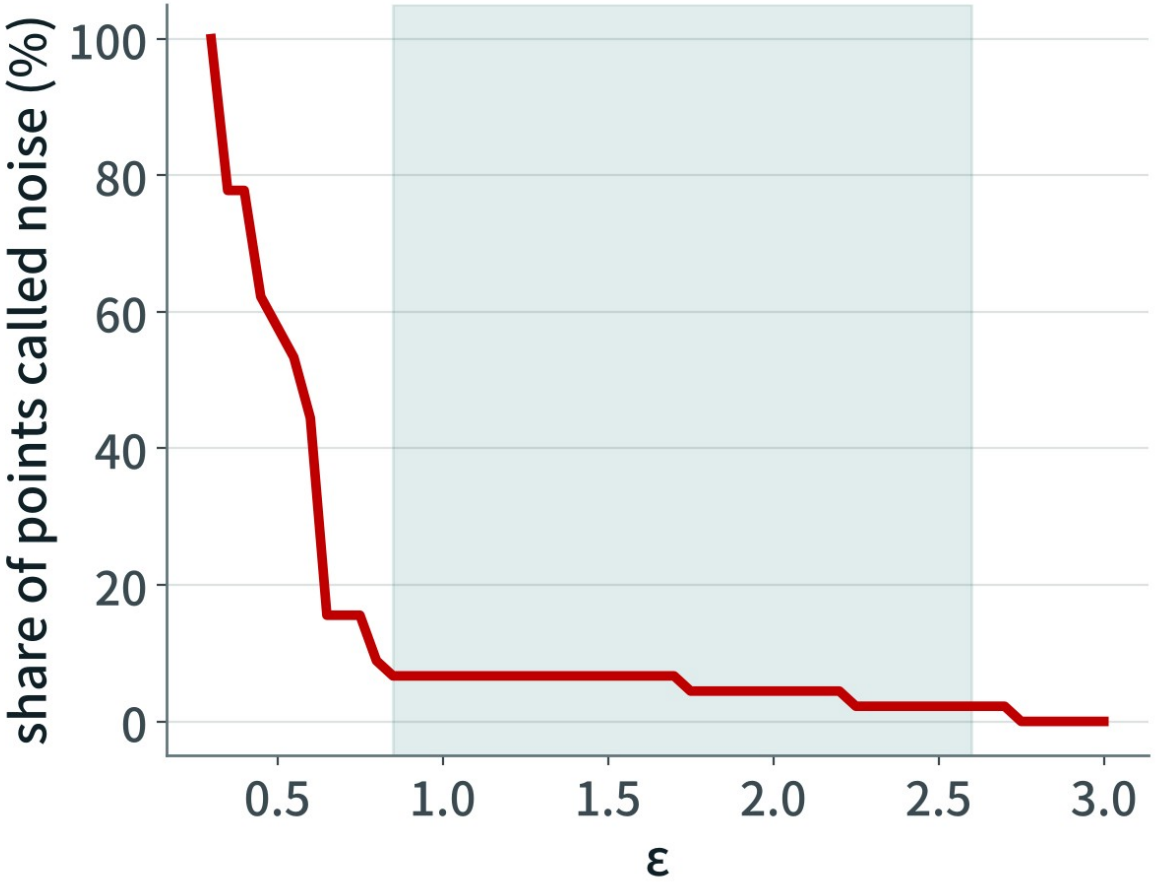
Any ε from 1.5 to 4.0 gives the same answer here.

Take the low end: a bigger ε risks merging separate clusters.

Choose ε inside the longest stretch where the answer does not change

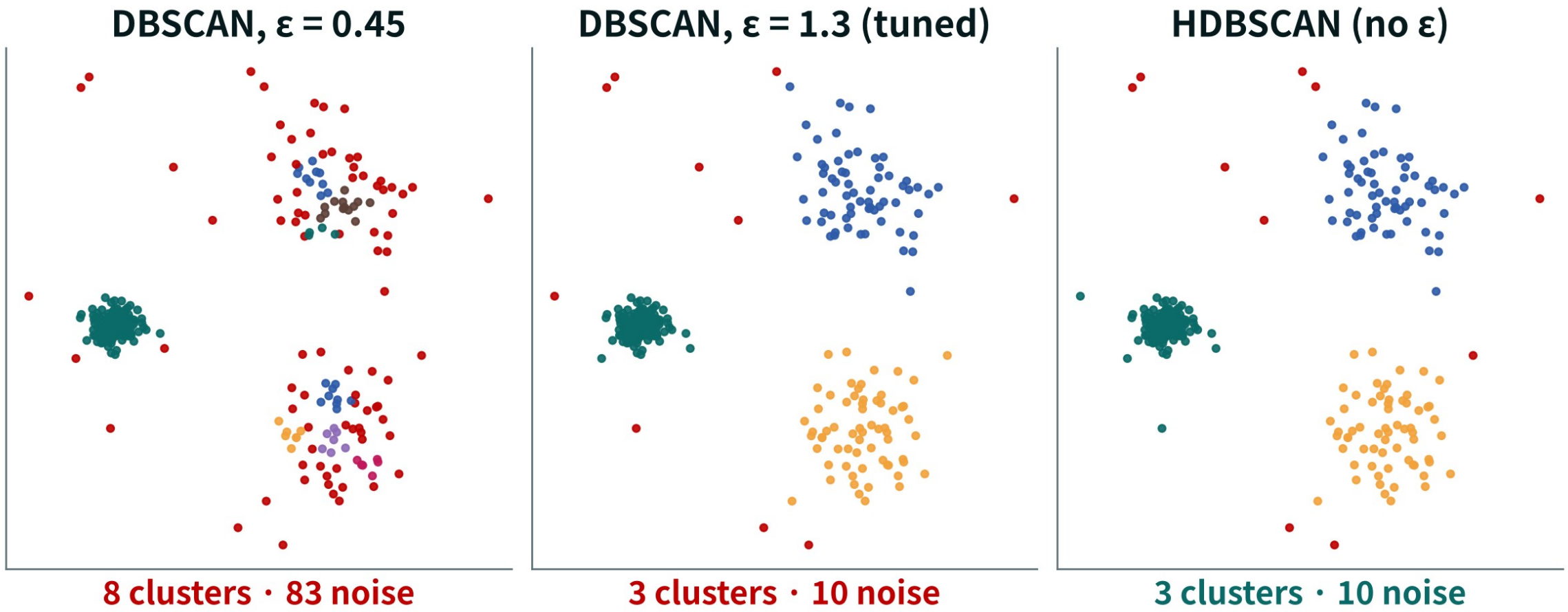
Main-deck data (45 points): DBSCAN run for every ε from 0.3 to 3.0





HDBSCAN tries every ε and keeps the clusters that last

One tight group and two loose groups. You only set the smallest group size (20)



Five ways to choose ε, from quickest to most careful

Standardise the features first, because ε is measured in distance units

Rule	What you do	Good for
Bend	Pick ε where the k-distance plot turns up	a quick first try
Flat stretch	Pick ε where the cluster count stays the same	checking the bend
DBCV score	Pick ε with the best density-cluster score	comparing settings
HDBSCAN	Skip ε; set the smallest group size	groups of mixed density
Business rule	Pick ε so noise is e.g. 1–5% of days	alerts you can handle

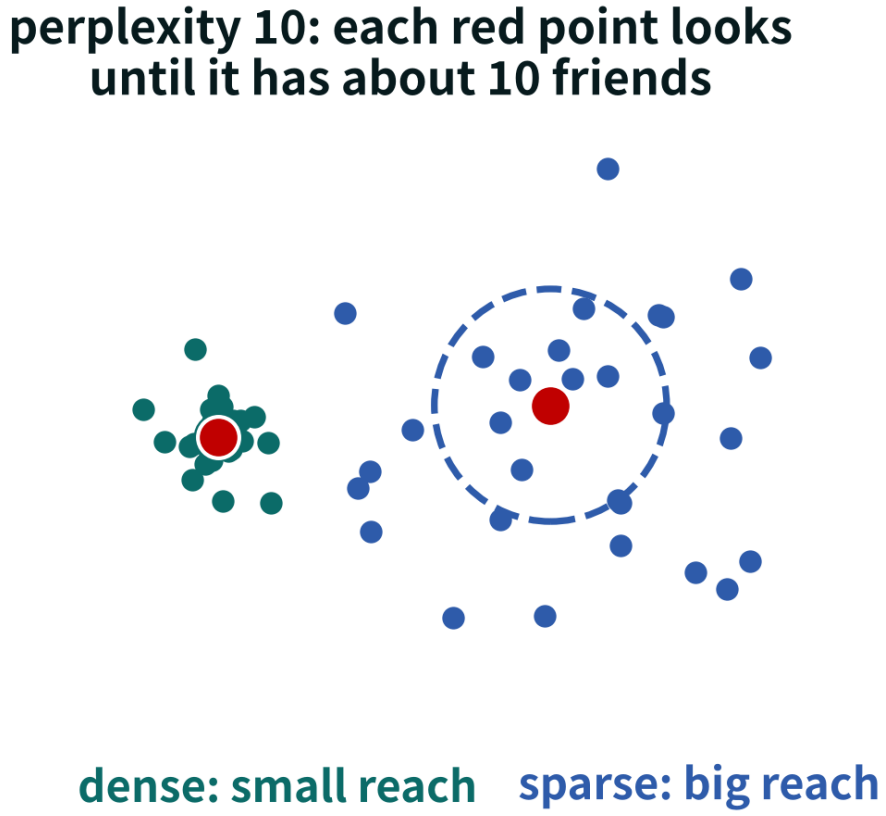
Perplexity sets how many friends each point looks for

t-SNE adjusts each point's reach so that it finds about this many neighbours

In plain words

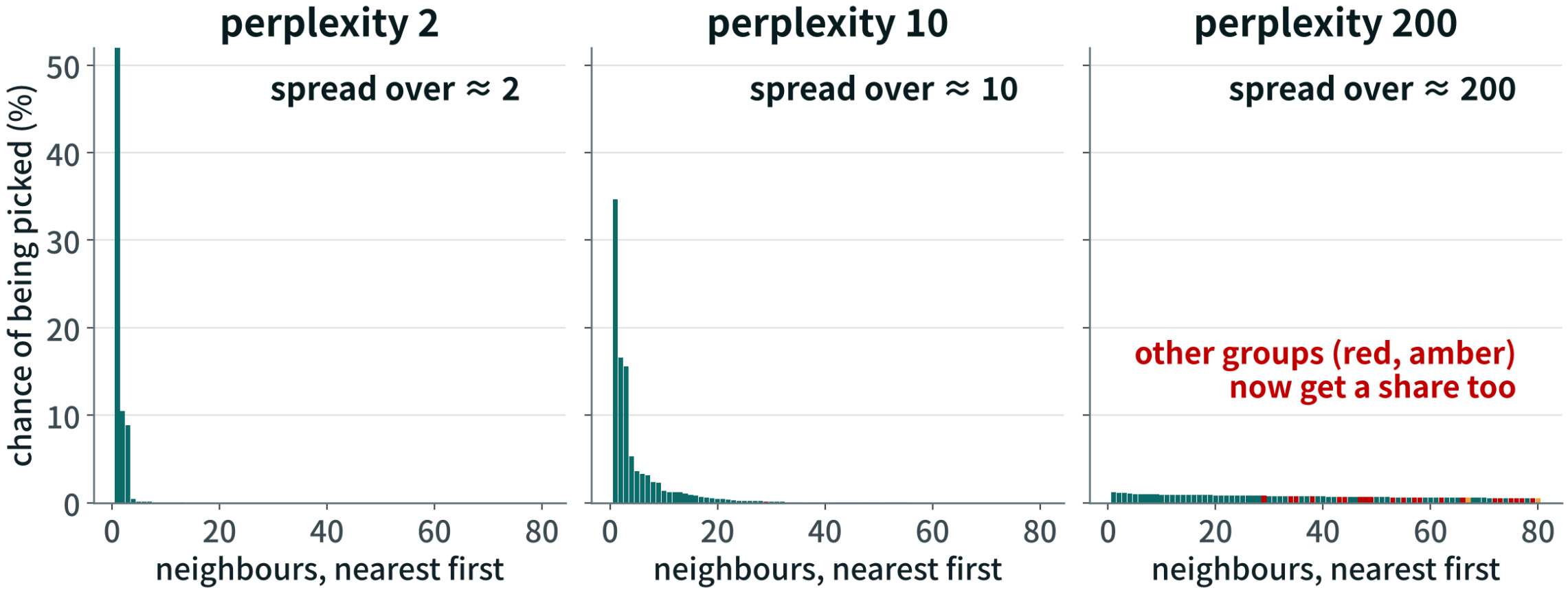
- ≈ friends per point
- Dense area: small reach
- Sparse area: big reach

Usual range 5–50



Perplexity becomes each point's chances of picking its neighbours

One of the 240 points; each bar is the chance of picking that neighbour, coloured by true group



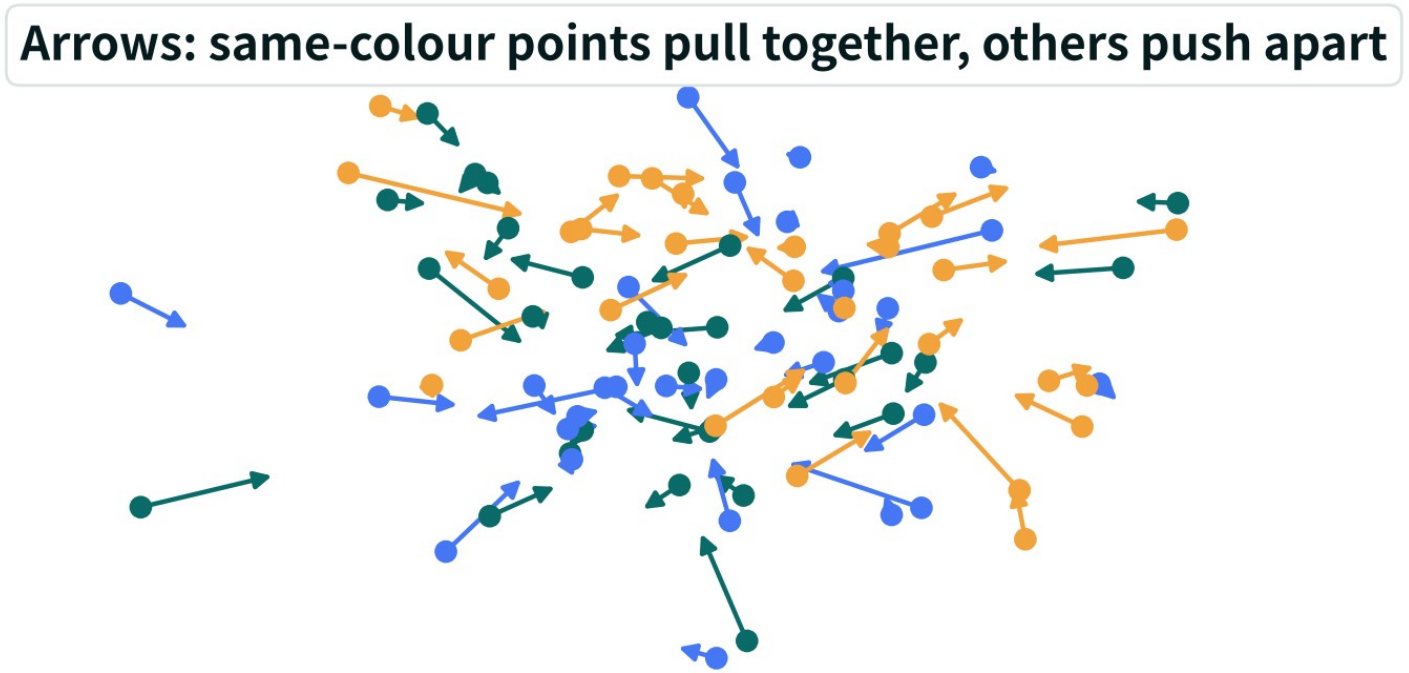
Each move pulls likely friends closer and pushes the rest away

All points start at random spots on the 2-D map and move at the same time

One move

- High chance: pull in
- Low chance: push away
- Every point at once

Repeat 500 times



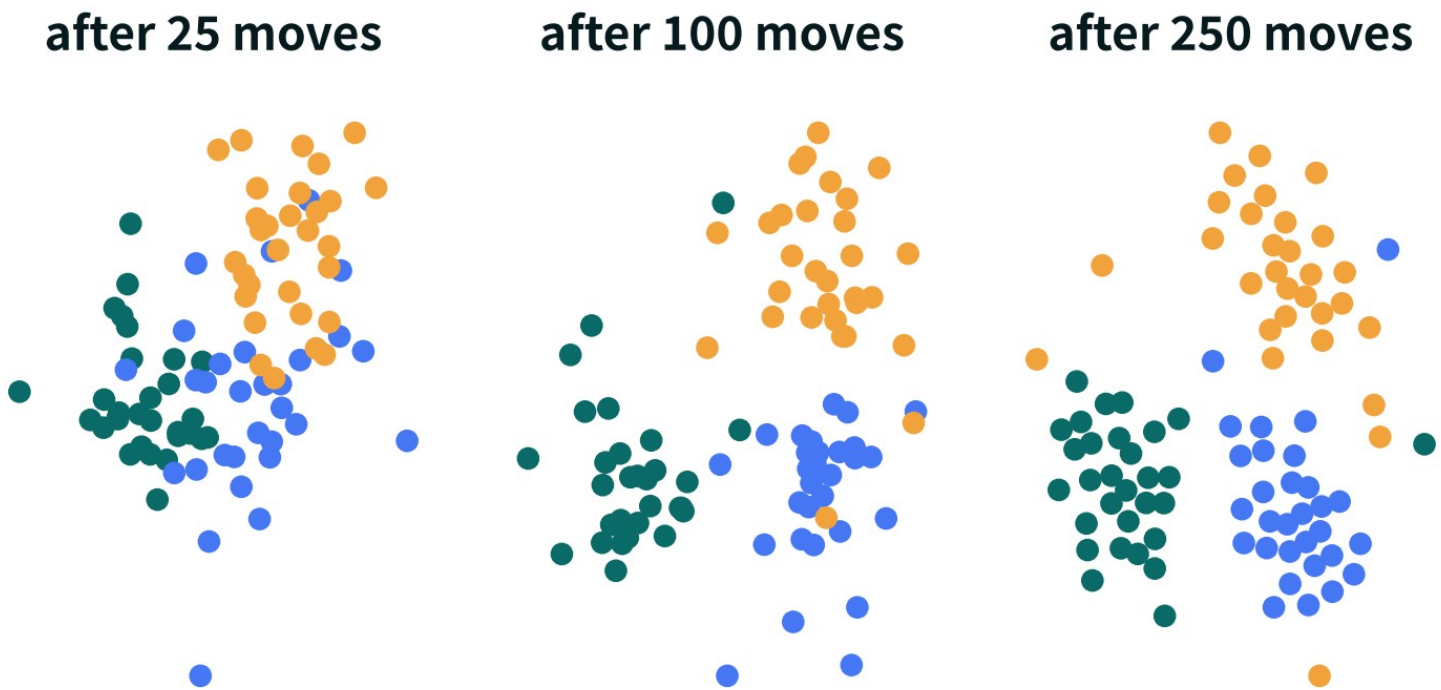
Friends of friends end up together, so groups appear on their own

There is no separate grouping step: islands form while the moves repeat

Why islands form

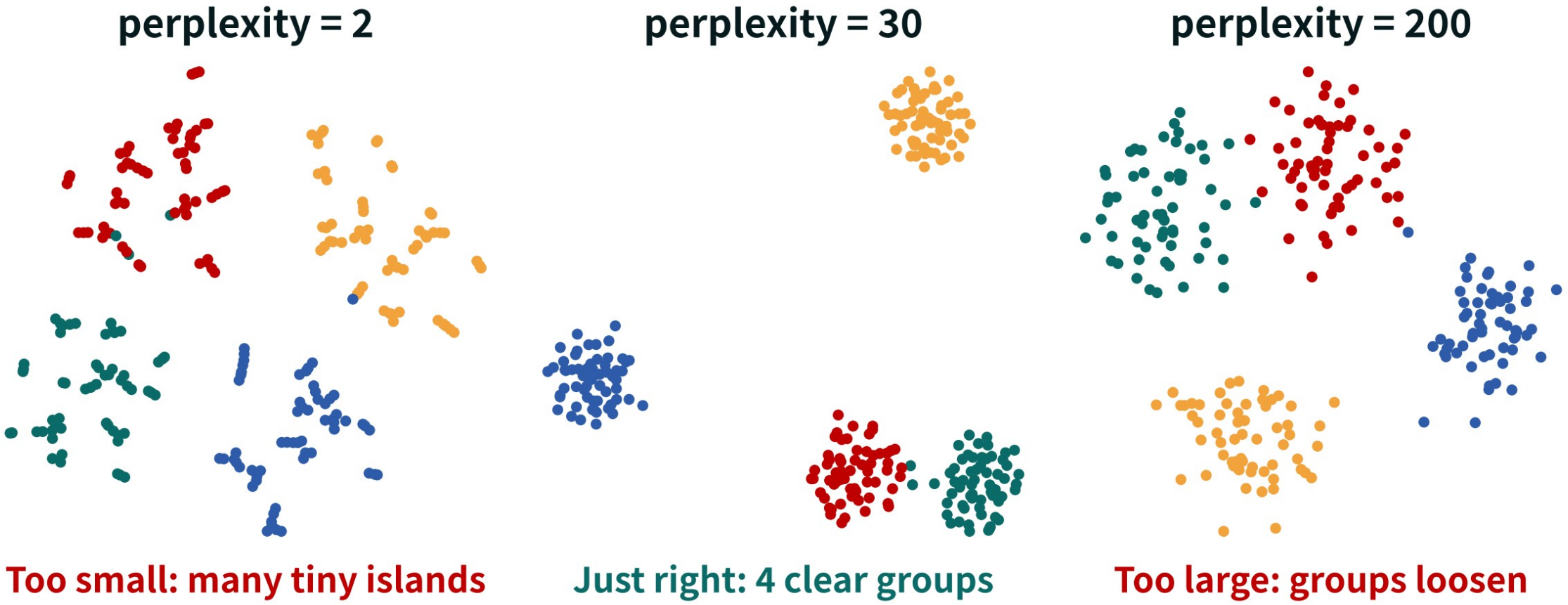
- A pulls B, B pulls C
- Chains close up
- Islands = groups

Groups emerge



Too small a perplexity shatters groups; too large a one loosens them

With 2 friends only pairs pull together; with 200, other groups pull too





Try It with Claude Code

One prompt reproduces every picture in this supplement

One prompt draws the k-distance plot, the ε sweep and HDBSCAN

Paste it into the Code tab, then read each figure and its one-line explanation

WHERE	INPUT	OUTPUT	FEEDBACK
Code tab	This prompt	4 figures	Ask why

[Input] Prompt · Claude desktop app, Code tab

Make 45 two-dimensional points: a curved arc of 26 points, a round blob of 16 points and 3 loners. 1) Draw the k-distance plot for minPts = 4 (each point's distance to its 3rd nearest neighbour, sorted) and mark the bend. 2) Run DBSCAN for ε from 0.3 to 3.0 and plot the number of clusters and the noise share against ε. 3) Make one tight blob and two loose blobs, and compare DBSCAN at a small ε with HDBSCAN (min_cluster_size = 20). 4) Make 4 groups of 60 points in 10 dimensions and draw t-SNE at perplexity 2, 30 and 200. Save and open every figure and explain each in one plain sentence.

ε: pick it at the bend.

Better: a flat stretch, or HDBSCAN.

Perplexity: friends per point.

Settings you choose by eye should be checked: change them a little and keep only what stays the same.